



This article features the most powerful and popular open source messaging and integration patterns server. Apache ActiveMQ is written in Java and comes with a full Java Message Service (JMS). It has high availability and high load scaling. It is used to reliably communicate between two distributed processes.

Every computer system frequently exchanges data, either locally, within the system or perhaps over the Internet. When we take a closer look, this is actually the data that is exchanged between processes running on the same machine or different machines. We call this data 'messages', which can vary in length and type. When two processes communicate with each other, it is a good idea to use a queue for messaging, because messages placed onto the queue are stored until the recipient retrieves them. This ensures robustness and reliability through persistence strategies. To maintain a high reliability level of the messages, message queues are used.

Today, there are a lot of message queues to choose from. In fact, there are dozens of message queues with different features and various pros and cons, and Apache ActiveMQ is one of them. Let's explore it.

What is ActiveMQ?

Apache ActiveMQ is a popular and powerful open source message-oriented middleware (software for communication between distributed applications). It supports the J2EE feature called JMS (Java Messaging Service). Apache ActiveMQ is released under the Apache 2.0 License. Apart from Java

Apache, ActiveMQ supports cross-language clients such as Perl, PHP, Python, C/C++, C#, Ruby and more. ActiveMQ is widely used with ESBs (Enterprise Service Bus) such as Apache ServiceMix, JBoss Fuse and Mule. It is also used with Apache Camel, a very powerful EIP (Enterprise Integration Pattern).

Features

Described below are a few of the features of Apache ActiveMQ:

1. Supports a variety of cross-language clients. Apart from the obvious Java support via JMS, it also supports Perl, PHP, Python, C/C++, C#, Ruby and more.
2. Full support for EIP (Enterprise Integration Pattern); Apache ActiveMQ is used with Apache Camel to support EIP.
3. Fully supports JMS 1.1 and J2EE 1.4.
4. RESTful API support — messaging directly from Web browsers using regular HTTP GET or POST.
5. Support for clustering; uses a network of brokers for scalability purposes.
6. Security can be customised for various types of authentication and authorisation.
7. Provides a wide range of connectivity options such as SSL, TCP, UDP, XMPP, multi-cast and more.
8. Easy for administration; does not require dedicated administrators. It provides very powerful monitoring tools.

Why use ActiveMQ?

Technologies using RPC (Remote Procedure Calls) are called tightly coupled distributed applications. Using RPC, one application can call the other application. There are